# PO: Optimization of Complex decisions

## Project: Optimization of the deployment of a carbon capture, transport and storage infrastructure

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## 1 Problem description

To reduce $CO_2$ emissions and meet our climate commitment, the carbon capture and storage (CCS) technology is more and more explored as a potential useful tool to mitigate global warming. Indeed, $CO_2$ is easier to capture when it is in high concentrations, like in the outputs of many industries. Moreover, it is likely that, despite their high levels of $CO_2$ emissions, some industries may be difficult to decarbonize and still necessary for our society. In those situations, building a CCS infrastructure offers a potential solution that researchers study.

In Europe, a few large capacity storage sites have been identified, but they are not numerous. In this project, we will aim at studying the development of a supply chain infrastructure, to route the $CO_2$ captured at emission sites towards storage sites. As transport is likely to play an important role in the development of such technology, in the rest of the subject we will use the term carbon capture, transport and storage (CCTS) to refer to this supply chain.

The goal of this project is to optimize the deployment of the $CO_2$ supply chain infrastructure over a time horizon of $T = 25$ years. To do so, we can use pipelines, rail and truck connections to transport the captured $CO_2$ to the storage sites. The deployment of such infrastructure should be carefully optimized so that it meets the predefined carbon emissions reduction scenario at minimum cost.

**Hint 1.** You will see that this problem can be relevantly modeled with a graph, with nodes and edges. Graphs are a convenient way to represent problems, and widely used in optimization. For example here, the edges would represent the transport infrastructure and the flow of captured $CO_2$ between geographical nodes. A classical notation convention when dealing with graphs is to use the notation $a_{ij}$ to index the arc connecting the nodes $i$ and $j$. If you do not

explicitly need both variables $a_{ij}$ and $a_{ji}$, you can use the convention $a_{ij}$ for $i < j$.

**Emission sites and carbon capture.** We assume we have a set $\mathcal{N}_C$ of potential capture sites, each site $i \in \mathcal{N}_C$ emitting a quantity $E_i$ of $CO_2$ each year. If we decide to implement it on node $i$, the carbon capture technology available enables us to capture 90% of the $CO_2$ emissions.

**$CO_2$ storage sites.** A small set $\mathcal{N}_S$ of storage sites are available, for which we assume they have unlimited capacity. The difference will come from their availability (some may be available right from the beginning of the planning horizon $t = 0$, while others may become available later), the geographical location and the transport infrastructures that are available to connect them.

**Transport hubs.** In our network, we add a small set $\mathcal{N}_T$ of transport exchange sites. They correspond to important nodes which are neither capture site nor storage site but can be used as transport hubs to connect different sites.

**Transport capacities.** The core of this project is to design the transport infrastructure of captured $CO_2$. Different transport options are possible, and we need to chose which of them are best to use.

- Truck connections. A truck connection is available between any pair of nodes in the network.
- Rail connections. We suppose the existing rail network serves some sites of the network, but not all. This information is given as input. We suggest to use the notation $w_{ij}^{rail}$ to represent whether or not a rail connection is available between nodes $i$ and $j$ ($w_{ij}^{rail} = 1$ meaning that a rail connection is available).
- Pipeline connections. A pipeline can be constructed to connect any pair of nodes in the network.

Each transport technology $l \in \mathcal{L}$ operates at a certain cost that depends on the amount of $CO_2$ transported and the length of the connection. For a certain technology $l$ and a certain connection of length $L$ between two nodes transporting a flow $F$ (the flow is the amount of $CO_2$ transported per year), at time $t$, the operating cost is calculated as the sum $C = \beta_{1,l} \cdot F \cdot L + \beta_{2,l} \cdot F$. Similarly, we should also account for the additional $CO_2$ emissions resulting from the chosen transport technology. On the same example as above, transport emissions are calculated as $E = \gamma_l \cdot F \cdot L$ (those emissions, which are the results of transporting $CO_2$, must be taken into account when making sure to respect the emissions reduction target).

**Pipeline connections.** Pipelines are a quite particular, heavy infrastructure, and the goal of this project is also to carefully model and optimize investments in such technology over a given time horizon $T$. One difficulty of the problem is that the capacity $S$ of the pipeline should be decided at the construction time $t$. For the sake of simplicity, we neglect construction delays, and consider that a pipeline can be constructed right from the year $t = 0$. However, after the construction, the pipeline capacity $S$ cannot be increased, meaning that the $CO_2$ flow transported by the connection cannot exceed $S$. As it is a heavy transport technology, we need to account for the associated investment costs. Investments costs should be annualized over the lifetime of the infrastructure, so that we can interpret the total cost without any bias coming from the choice of the time horizon $T$. In practice, this means that a cost $C = \alpha_1 \cdot S + \alpha_2 \cdot L$ will be paid each year the pipeline is in service for a connection of length $L$ and of capacity $S$.

**Emissions reduction scenario.** Eventually, our deployment strategy should be optimized regarding our climate targets. This is a constraint that will be imposed to our optimization program. The way to proceed is to progressively reduce yearly emissions, until the target is met by the end of the planning horizon, i.e., at $T$. Let $Q_{init}$ be the total yearly emissions at the beginning of the project. The objective is to reach the final yearly emissions of $Q_T$ at minimum cost while respecting (at least) a linear decrease of emissions over the years.

**Optimization objective.** The result of the optimization program should give us a deployment plan, where we know year by year 1) when the $CO_2$ capture technology should be installed at each capture site, 2) how much $CO_2$ is transported on each connection and for each technology. To be valid, the CCTS plan should also make sure to respect the mass balance constraint: as $CO_2$ is transported over the network, from node to node via transport connections, the transport planning must not have any "leak" (i.e., the flow arriving at a node that is not a storage site should be equal to the departing flow from that node). The goal is then to optimize the deployment strategy so that the $CO_2$ emissions reduction constraint is respected at minimum cost.

# 2 Expected work

You are expected to propose a mathematical model to solve this optimization problem. Several instances will be provided, as well as with the numerical parameters. You can solve it with the tools of your choice. A critical analysis on the solving algorithm, its performance and the difficulty of the problem is also expected. At the end of the project, you should produce two outputs. A report should summarize your work, and explain how you modeled the problem, how you solved it, and present your analysis. To guide you in your work and the writing of your report, you may use the questions below as some indications

on the expected content. Moreover, you should also produce a notebook where you present your code and how you obtained your numerical results.

### 2.1 Q1.

Propose an optimization model to compute a deployment planning that minimizes the total cost while respecting the constraints.

### 2.2 Q2.

Solve the problem for the 3 instances and using the provided numerical parameters. Pay attention to the outputs and information provided by the optimization solver: number of variables and constraints, solving time, gap, explored nodes (if branch-and-bound algorithm), etc..., they will help you understand the solving performance.

**Hint 2.** When you get a solution, it is very important to check it. First, you should make sure that the constraints are respected. Second, you are encouraged to think of a way to validate that the obtained solution is correct (it often happens that we make errors in the modeling and/or in the code).

### 2.3 Q3.

Now that you have a good model and a working code, you can study the sensitivity of the solution and the solving performance to a couple of parameters (but not $Q_T$, which is kept for the last question). To do so, select one or two parameters that you suspect they could influence the solving difficulty, and run your optimization program for different values of those parameters. By studying the outputs of your optimization solver, conclude on the respective influence of those parameters.

### 2.4 Q4.

Often, in this type of problem, two different objectives are conflicting: we would like to reduce as much as possible the emissions while spending as less money as possible. What can you do to compute a Pareto front ? Your role, as an engineer, is to provide accurate information to guide the decision-maker in its carbon policy. If you sum the emissions over the entire period, you can compare it to the $CO_2$ that would have been emitted without any carbon capture and storage policy over the same period. This allows you to compute the cost per unit of $CO_2$ not emitted. Plot it for different scenarios of emissions reduction.

## 3 Instances - numerical values

Instances (describing the configuration of the network, availabilities of transport technologies, etc...) and numerical parameters will be provided in a separated

file.